NewsLens: An Explainable News Classification System

Course Project Proposal | Track B: Demonstration

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Project overview

A useful news-routing tool needs more than a predicted topic. Readers should be able to inspect which text features influenced the decision and recognize when a prediction merits manual review. This project studies a compact, locally runnable solution for English sports, politics, and business news. It treats probability quality and explanation validity as separate questions from classification accuracy.

The representation experiments from Homework 1 supply controlled baselines. The course project extends those baselines into an interactive system with calibrated decisions, selective prediction, exact linear explanations, and explicit verification. The aim is a working demonstration with measured behavior, rather than a larger model or an API wrapper.

Problem definition and research questions

Input: one English news article. Output: class probabilities, a preferred topic, the strongest terms supporting it over the runner-up class, and an accept-or-review decision. The three labels are closed-set topics; the output is not a fact-checking verdict.

RQ1: How do lexical, averaged word-vector, and pretrained neural representations compare under a shared test split? RQ2: Does temperature scaling improve held-out negative log likelihood and calibration error? RQ3: How does a review threshold change accepted coverage and accuracy? RQ4: Can the displayed term contributions exactly reconstruct a class-score margin, and how sensitive are predictions to removing those terms?

System design and original implementation

The data layer preserves the supplied CSV files, their checksum, and fixed row identifiers. The model layer trains Binary BoW, Word Frequency, TF-IDF, GloVe-100d, two 100-dimensional Word2Vec variants, and the required BERT-base model for offline comparison. Vocabulary, IDF, NYT embeddings, and scalers are fitted on the training partition only.

The decision layer divides the validation set into disjoint temperature-fitting and threshold-selection subsets. The explanation layer computes $x_j(w_{aj} - w_{bj})$ for the predicted class a versus runner-up b and verifies that all terms plus the intercept reconstruct the score gap. A local Flask interface integrates input validation, model selection, prediction, explanations, review guidance, and recorded experiment results.

Data and evaluation

The supplied NYT corpus contains 11,519 labeled articles; AG News supplies 90,000 texts for one Word2Vec experiment. A seed-42 shuffle yields 9,215 training, 1,152 validation, and 1,152 test rows. Every method uses the same saved split. Exact duplicate text and class imbalance are audited, with a supplementary test result excluding articles repeated verbatim in training.

Classification is evaluated with accuracy, macro-F1, per-class precision/recall/F1, and confusion matrices. Confidence is evaluated by negative log likelihood and ten-bin expected calibration error. Selective prediction reports both coverage and accepted accuracy. Explanation tests measure maximum score-reconstruction error and compare deleting the top five positive contributors with deleting the same number of random present features.

Expected outcomes and feasibility

The expected deliverables are a runnable local demo, full source code and dependency information, test-row predictions and metrics, an ACL-style eight-page final report, slides, and a contribution statement. The project tests a hypothesis about practical transparency; it does not assume that BERT or calibration will necessarily improve every metric.

The implementation uses established open-source libraries for standard algorithms while adding its own data contract, decision policy, explanation logic, system integration, and tests. The local computer has an i7-13700H CPU and 16 GB RAM, with no detected NVIDIA GPU. Neural fine-tuning may require a long CPU run, but the deployed demo can use a compact lexical model. Negative or inconclusive results will be reported directly.

Member responsibilities and timeline

Yi Tan is the sole member, responsible for requirements, data preparation, modeling, implementation, testing, analysis, writing, and presentation. Codex assists drafting and execution; the final contribution statement will distinguish student work from libraries, pretrained weights, and AI assistance. The student must review and be able to explain the final materials.

Stage	Planned work and acceptance criterion
Proposal | 12 Oct 2026	Define the problem, architecture, evaluation plan, and feasibility; submit this PDF.
Midterm | date to be announced	Show a working system core, real initial results, a 2–3-page report, and up to five slides.
Final presentation | date to be announced	Finish comparison and system evaluation; prepare an eight-minute talk and live demo.
Final submission | date to be announced	Submit the ACL-style report, slides, source, dependencies, demo, and contribution statement.

References consulted for the proposal

Devlin et al. 2019, BERT, ACL Anthology N19-1423; Pennington et al. 2014, GloVe, ACL Anthology D14-1162; Mikolov et al. 2013, arXiv:1301.3781; Guo et al. 2017, On Calibration of Modern Neural Networks, PMLR 70.